# ALL ABOUT CIRCLES

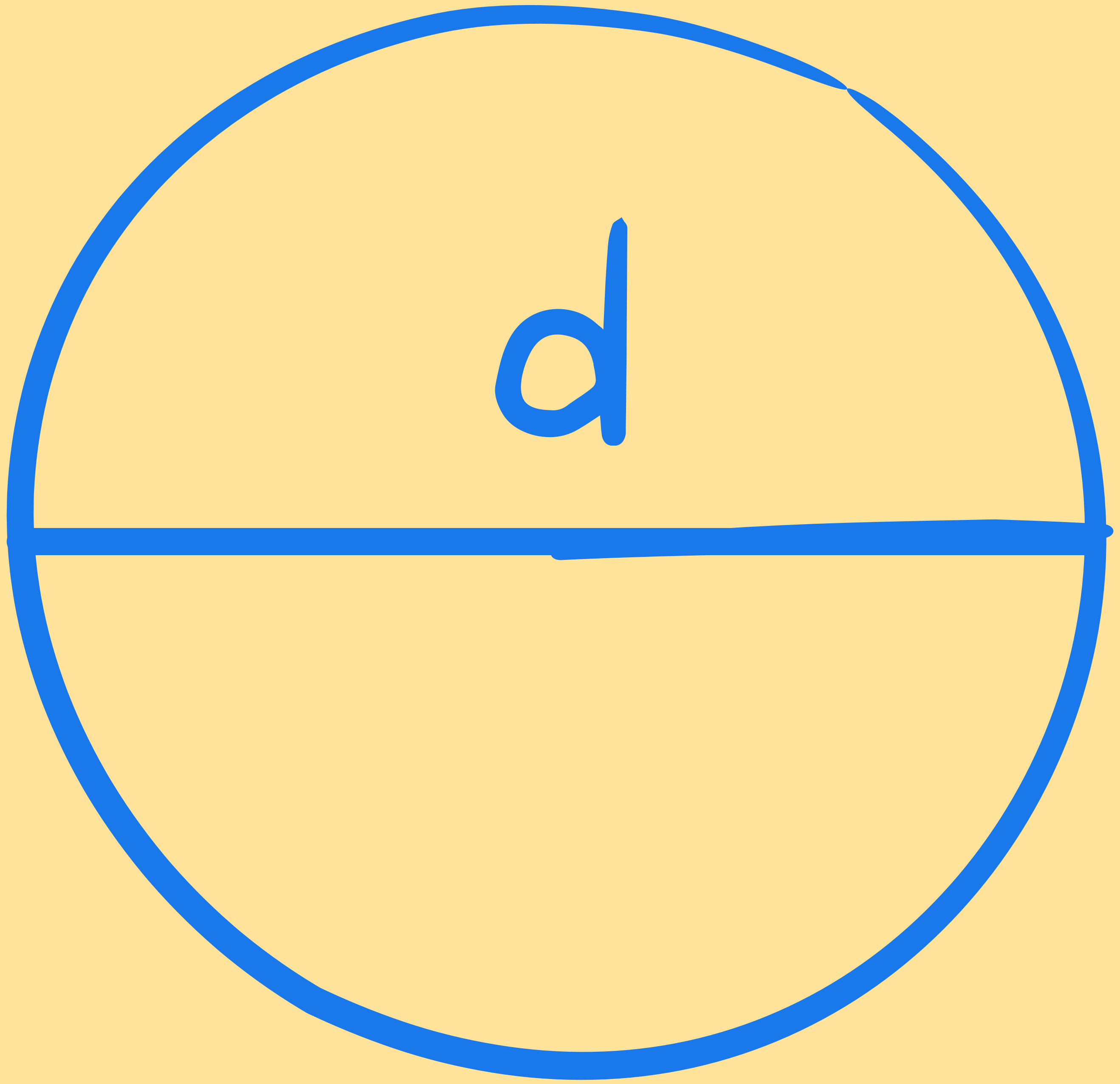
MATHS PRESENTATION

# CIRCLES ARE SUPER COOL SHAPES!

Let's see why.

## R = RADIUS

The distance from the centre of a circle to any point on its curved edge.

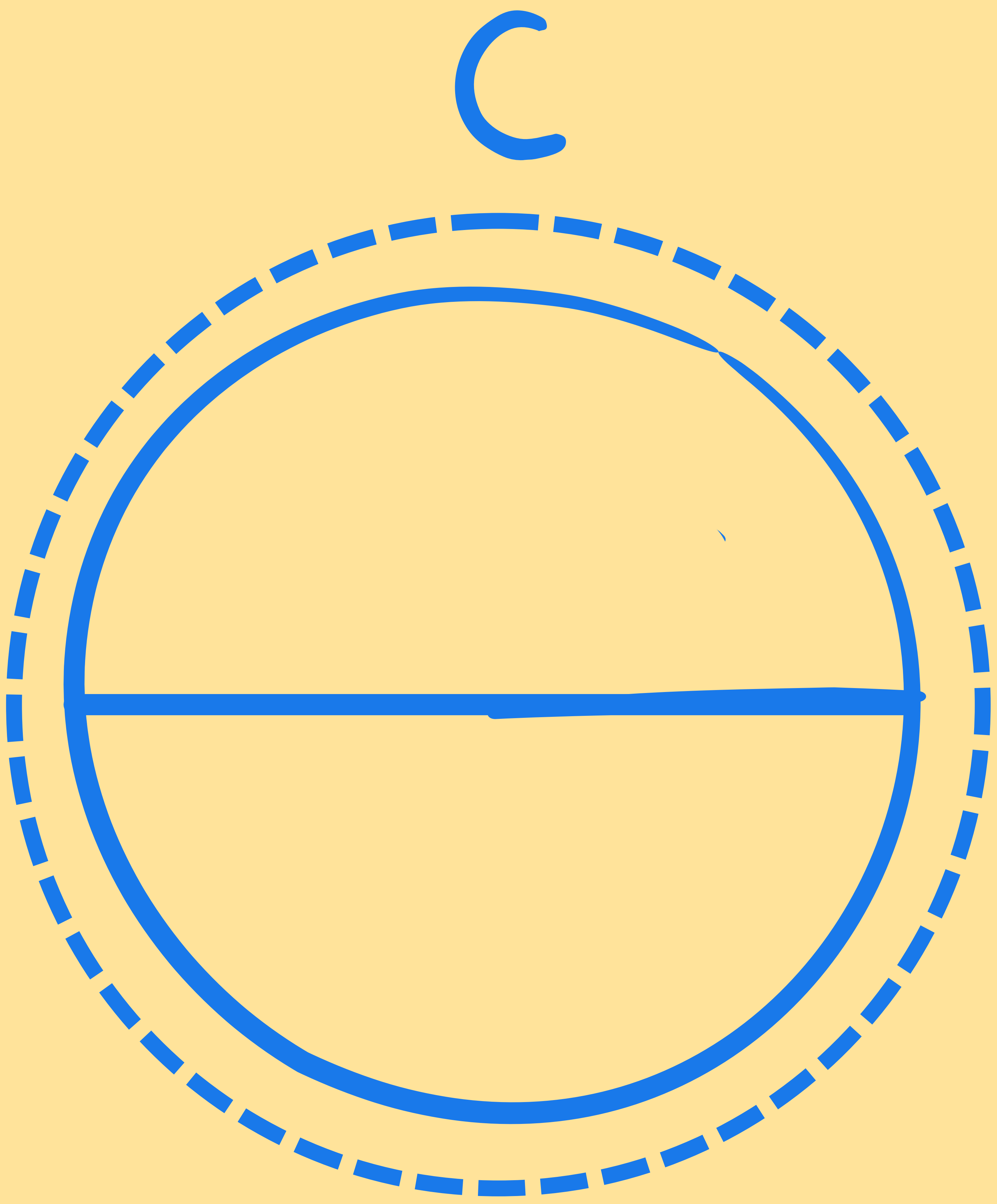



## D = DIAMETER

The distance across a circle through the centre.

## C = CIRCUMFERENCE

The distance around the outside of a circle.




THERE'S A SPECIAL NUMBER COMMON TO ALL CIRCLES.

THIS NUMBER IS PI.

PI IS A NEVER-ENDING DECIMAL NUMBER.

3.1415926535 8979323846 2643383279 5028841971 693...

The symbol we use for pi is $\pi$

$$\text{pi} = \frac{\text{circumference}}{\text{diameter}}$$

THE **CIRCUMFERENCE DIVIDED BY THE DIAMETER OF A CIRCLE** IS ALWAYS PI, NO MATTER HOW LARGE OR SMALL THE CIRCLE IS!

WE CAN USE THIS INFORMATION TO HELP US SOLVE PROBLEMS WITH CIRCLES!

# CIRCUMFERENCE


Circumference Formula:

$$C = 2\pi r$$

# AREA OF A CIRCLE

Circle Area Formula:

$$A = \pi r^2$$

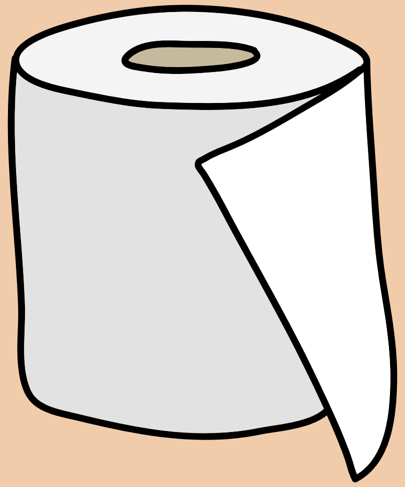
# What are 3D shapes?

3D shapes, or three-dimensional shapes, are objects that have length, width, and height.

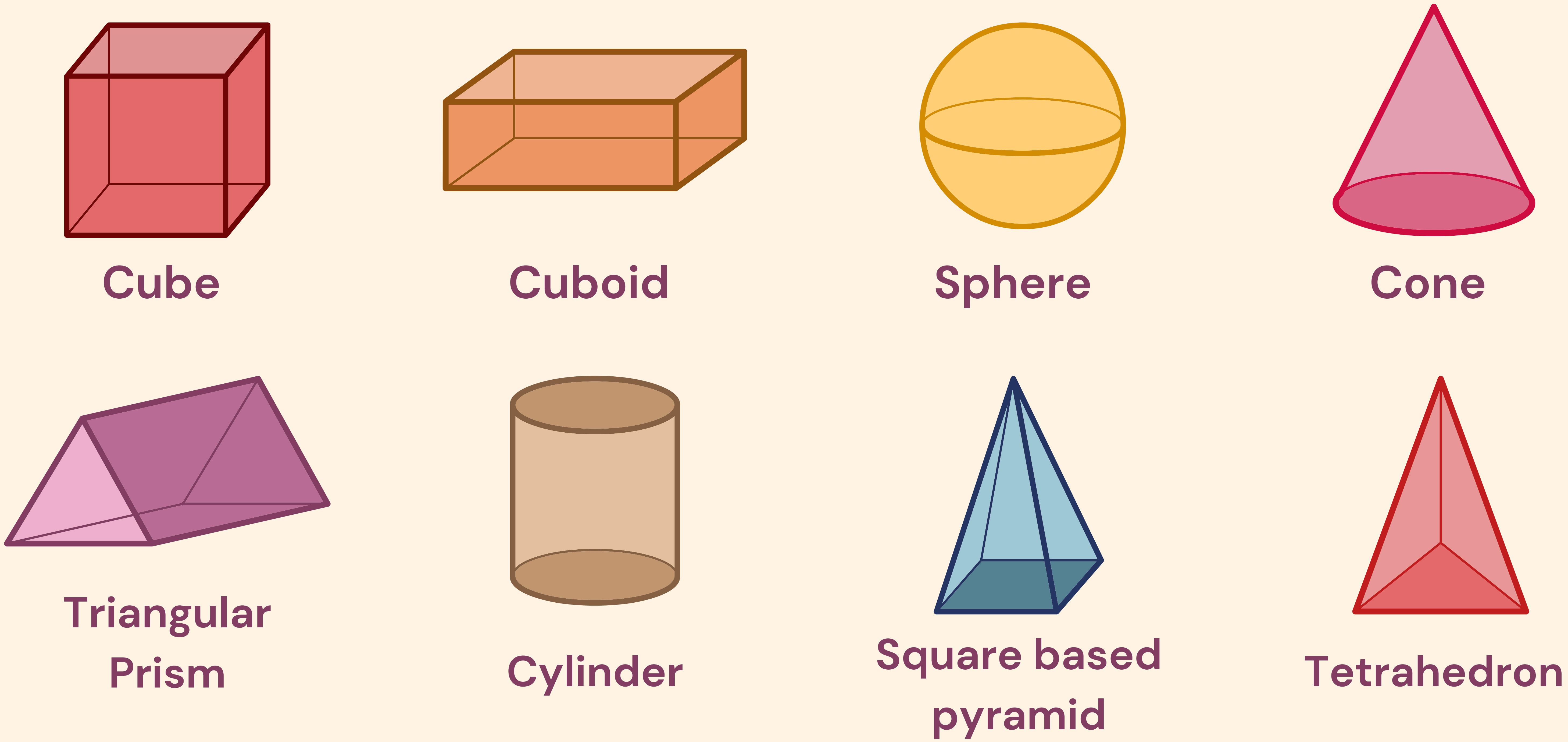
We will explore the mathematical names of 3D shapes and their properties.

## Real life objects:



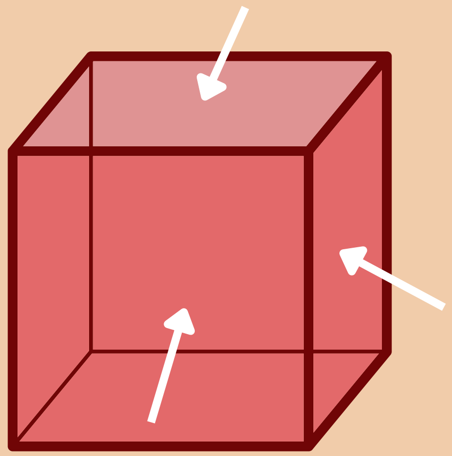



# Common 3D shapes



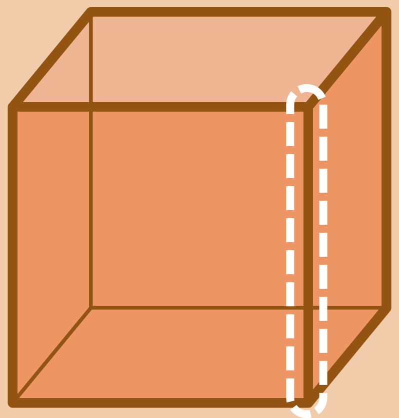
# Properties of 3D shape

## Faces



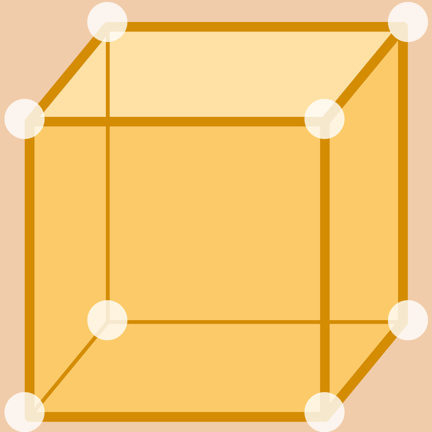
Faces are flat or curved surfaces on a 3D shape

## Edges



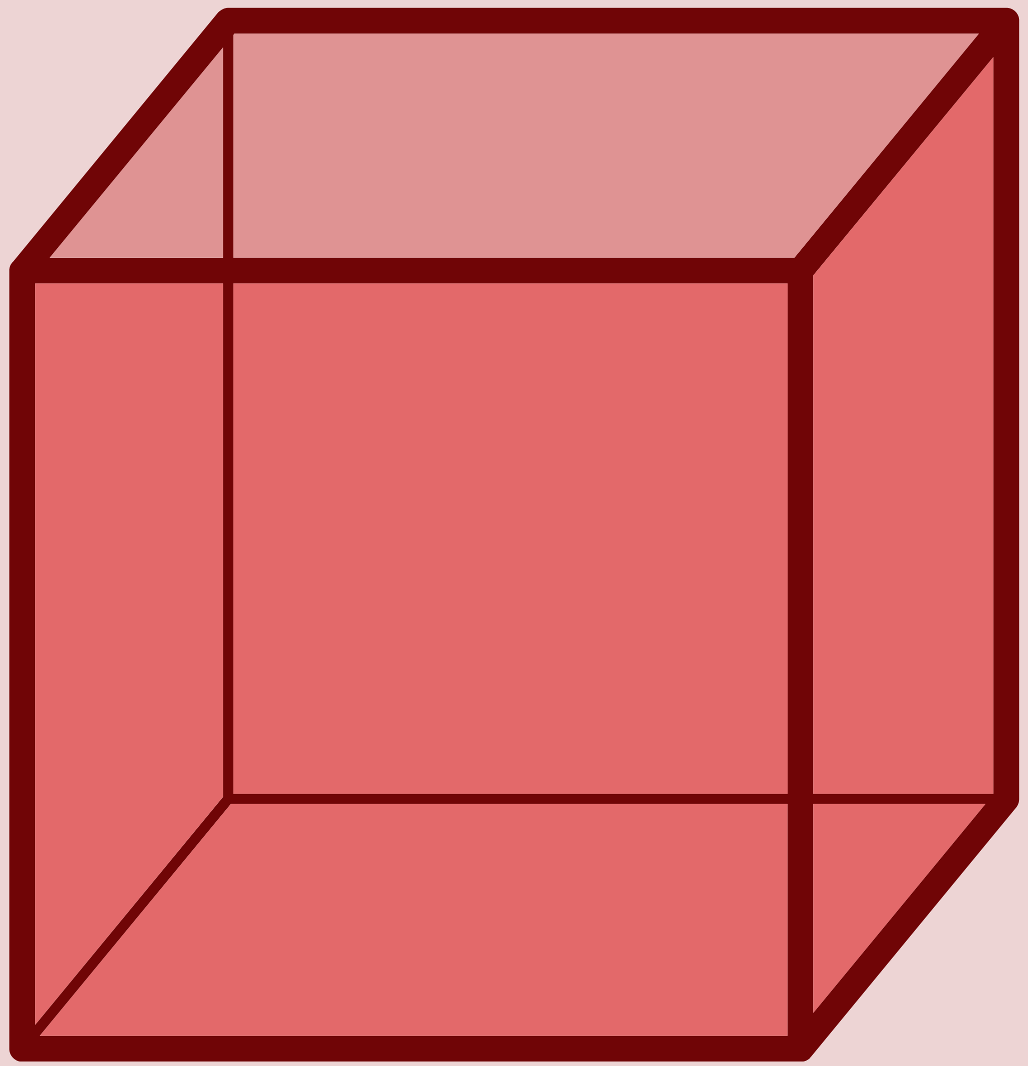
An edge is where two faces meet

## Vertices



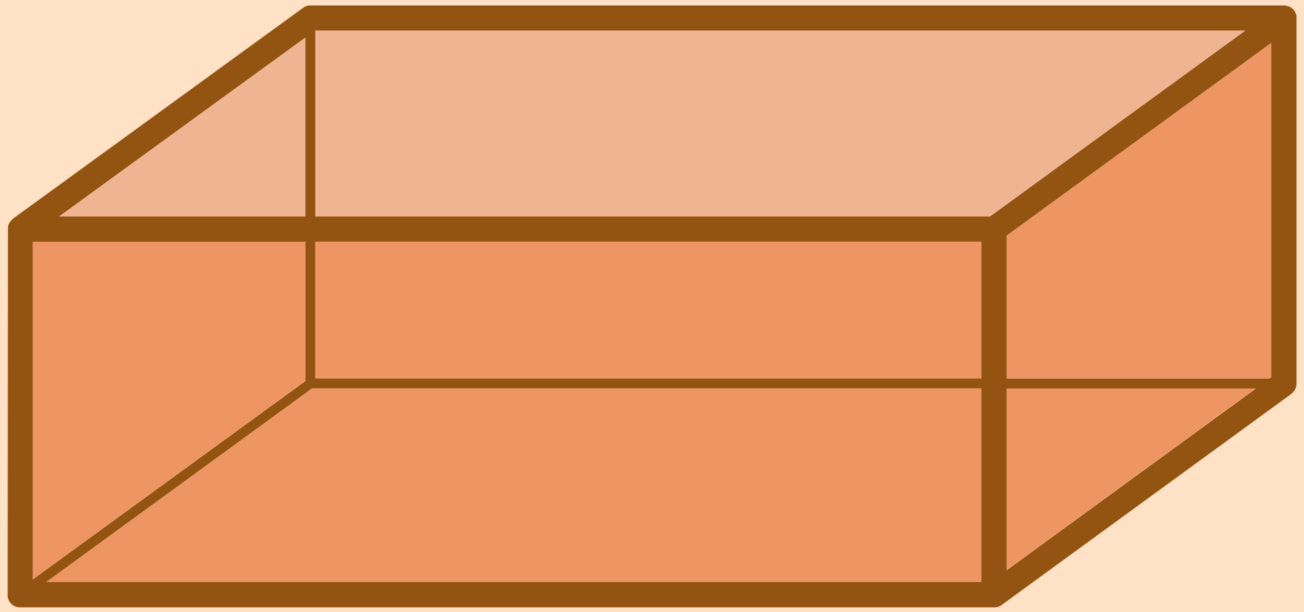
Vertices/corners are where two or more edges meet

# Cube



| | |
|---|---|
| Faces | 6 |
| Edges | 12 |
| Vertices | 8 |

# Cuboid

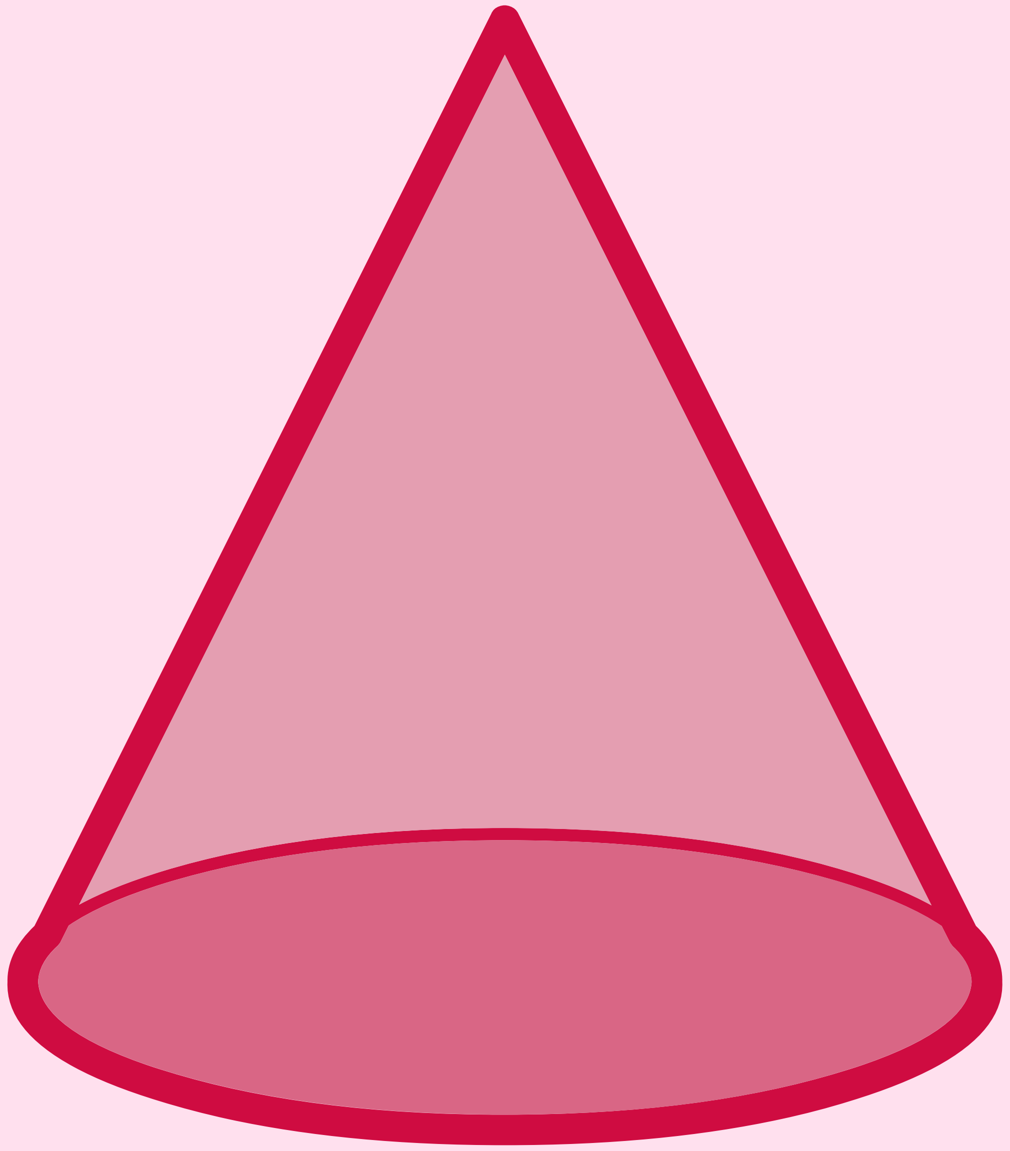


| | |
|---|---|
| Faces | 6 |
| Edges | 12 |
| Vertices | 8 |

# Sphere

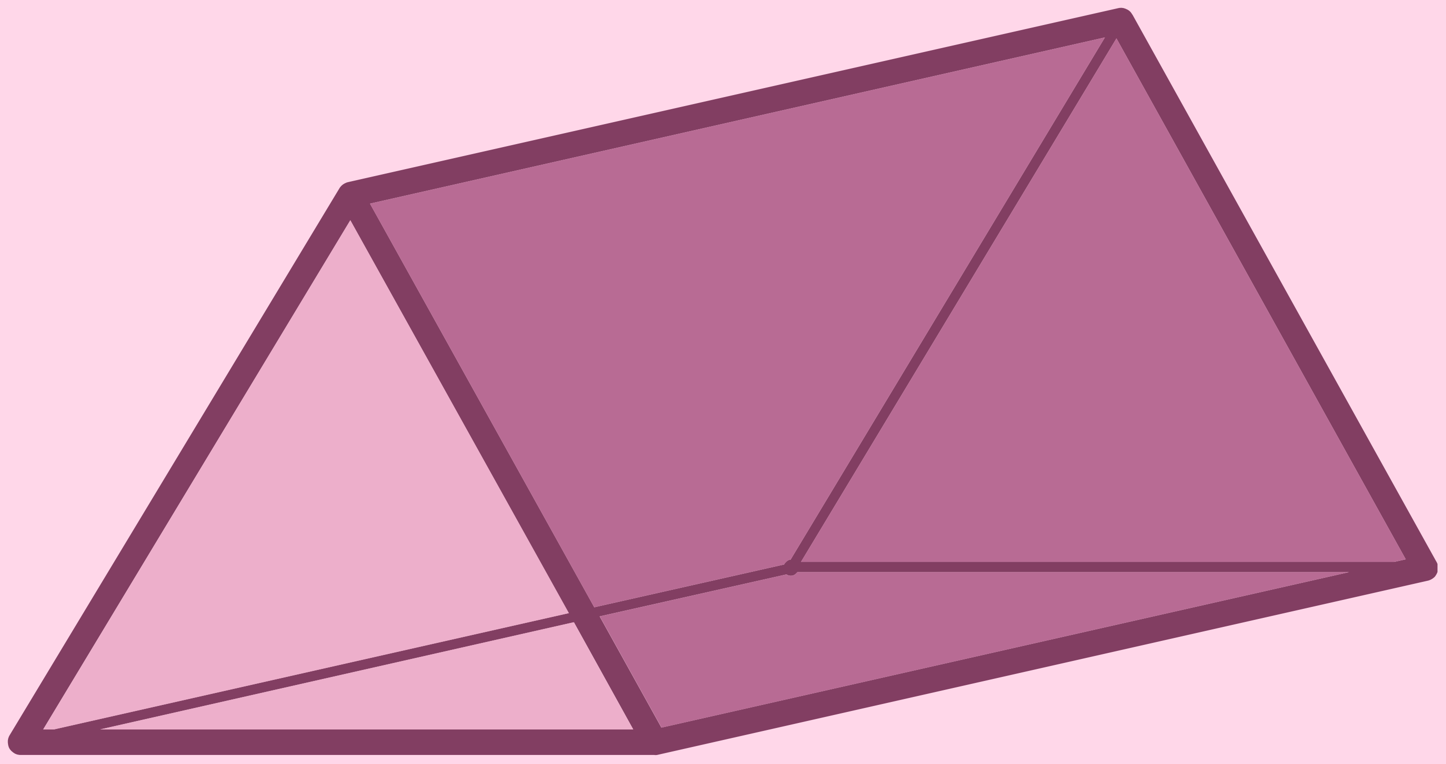
| | |
|---|---|
| Faces | 1 |
| Edges | 0 |
| Vertices | 0 |

# Cone



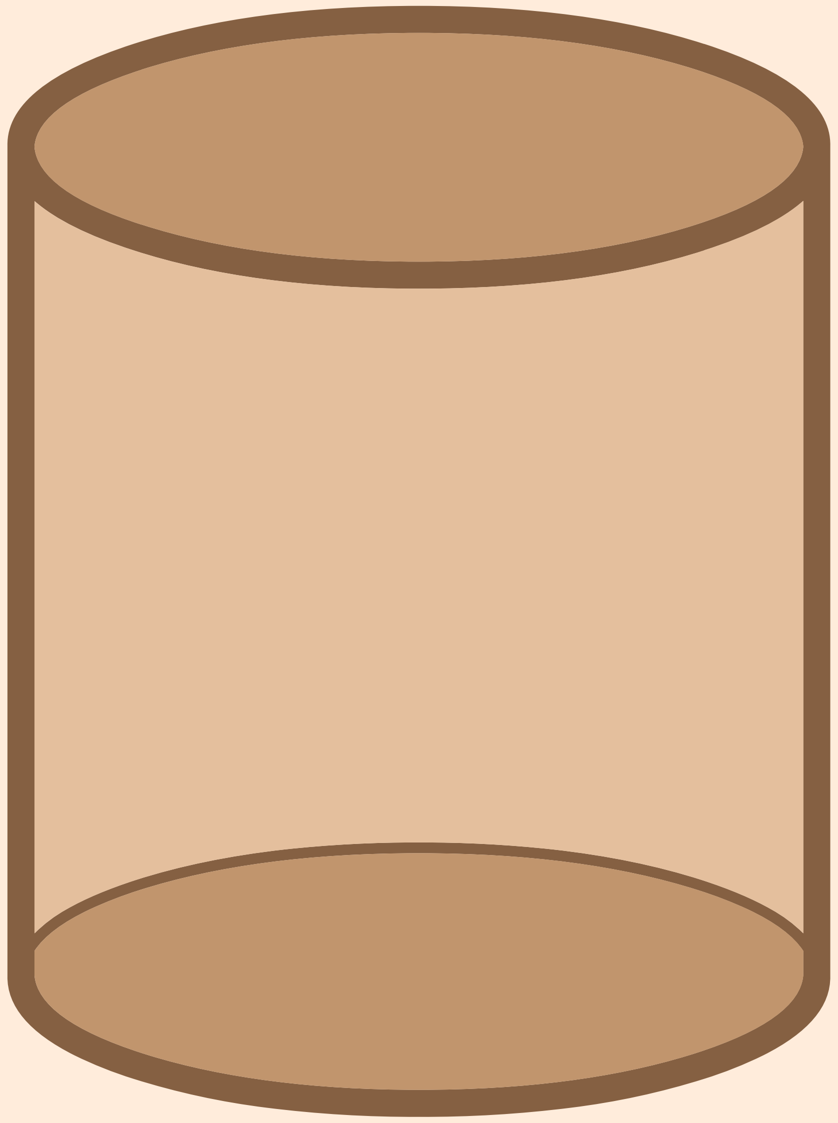
| | |
|---|---|
| Faces | 2 |
| Edges | 1 |
| Vertices | 1 |

# Triangular Prism



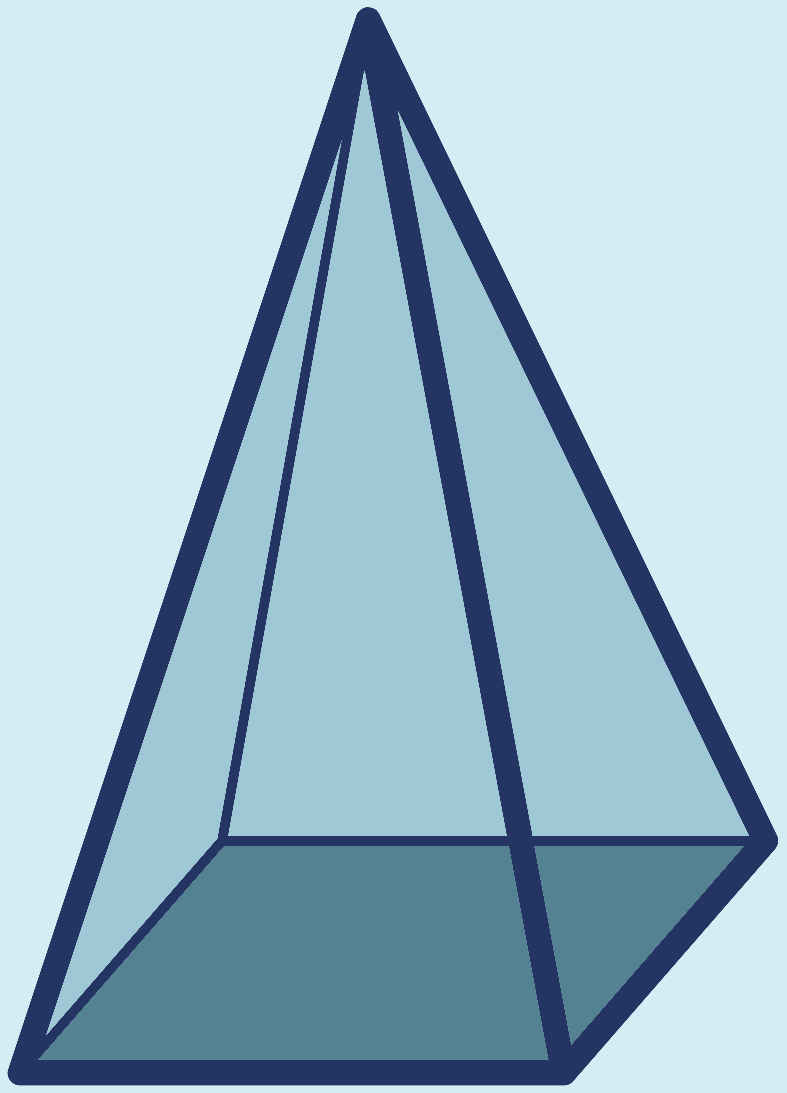
| | |
|---|---|
| Faces | 5 |
| Edges | 9 |
| Vertices | 6 |

# Cylinder



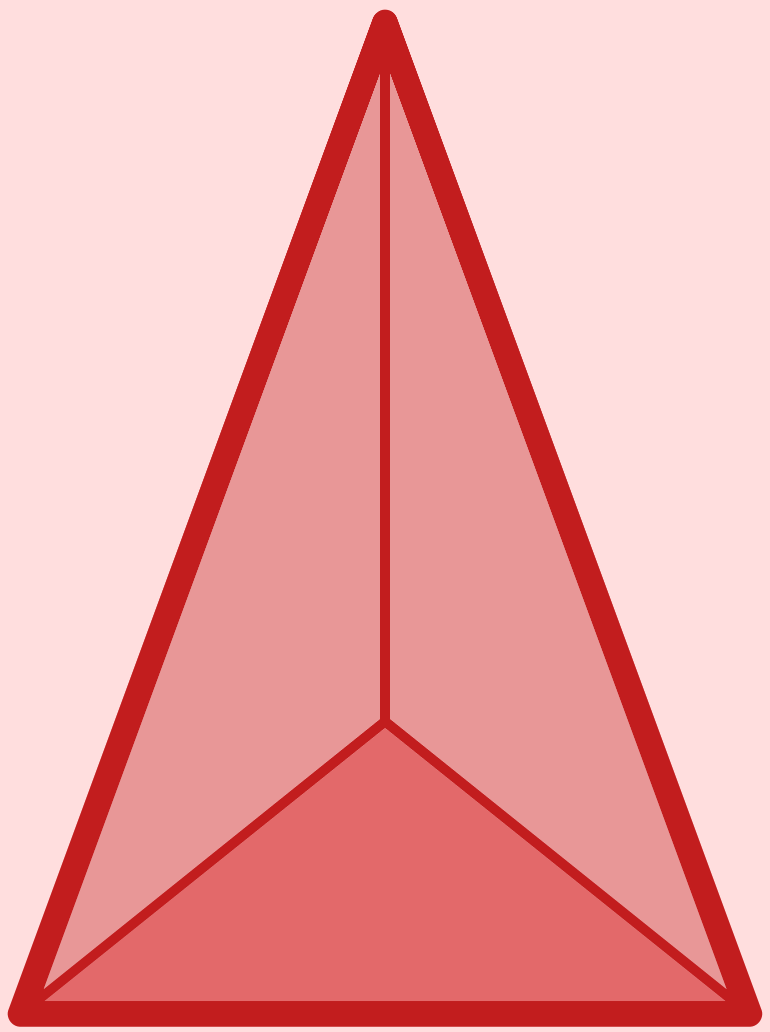
| | |
|---|---|
| Faces | 3 |
| Edges | 2 |
| Vertices | 0 |

# Square-based Pyramid



| | |
|---|---|
| Faces | 5 |
| Edges | 8 |
| Vertices | 5 |

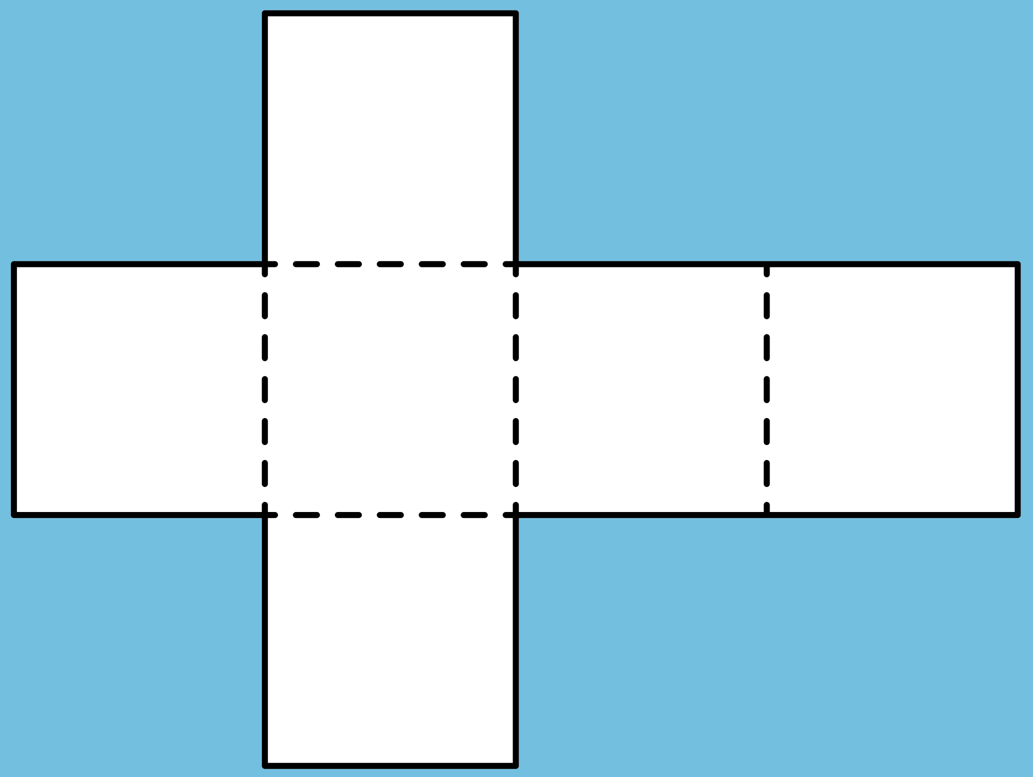
# Tetrahedron



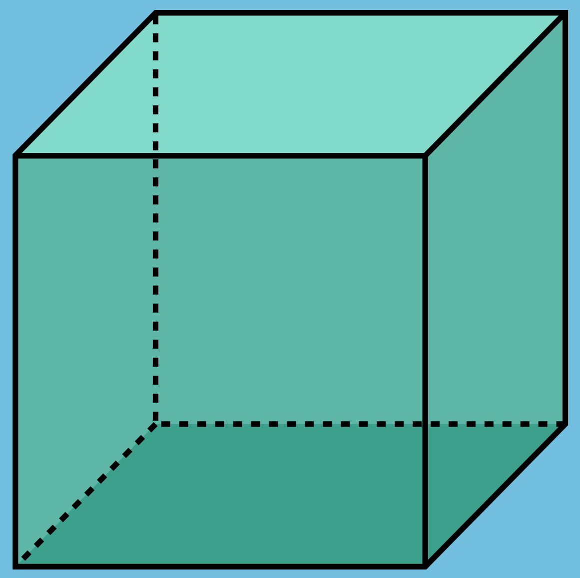
| | |
|---|---|
| Faces | 4 |
| Edges | 6 |
| Vertices | 4 |

# WHAT ARE NETS?

The net of a 3D shape shows what it would look like if it was unfolded and laid flat. A net can be reassembled to create a 3D shape.



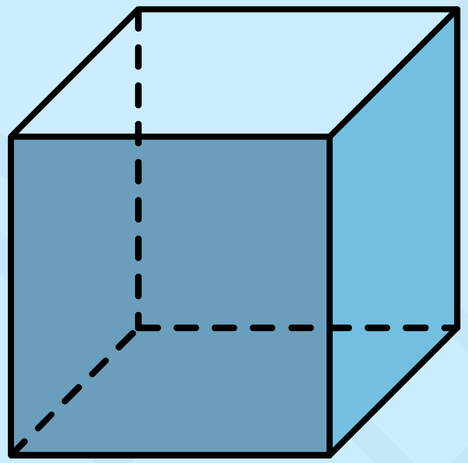
net of a cube

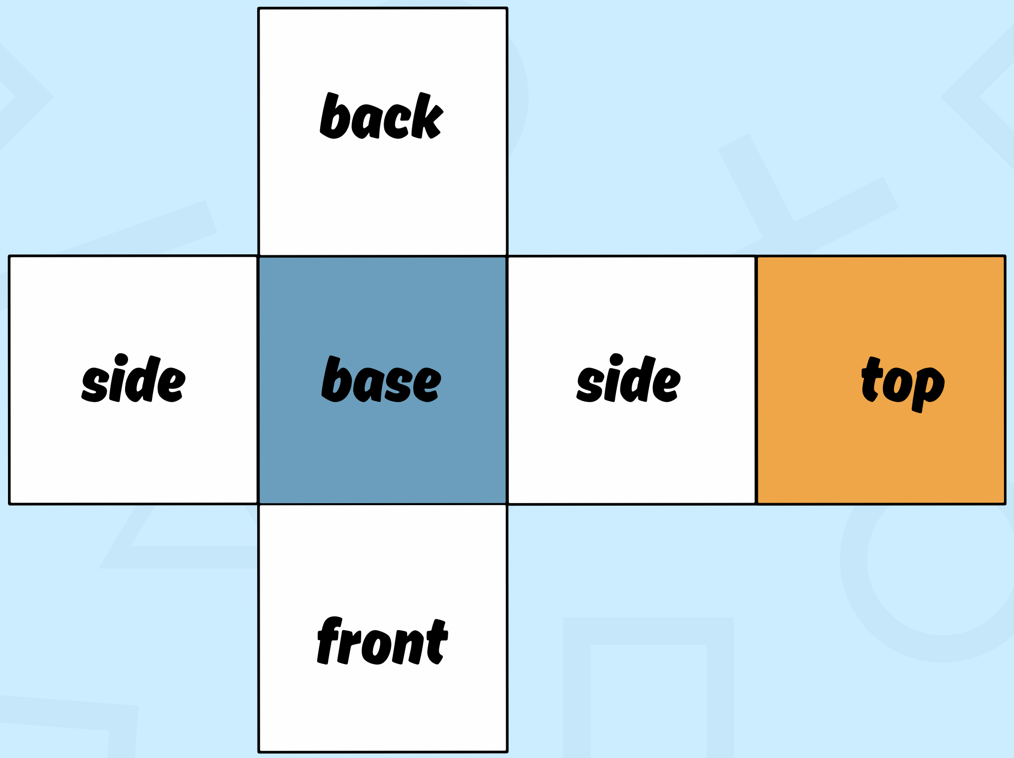


cube

# Net of a Cube

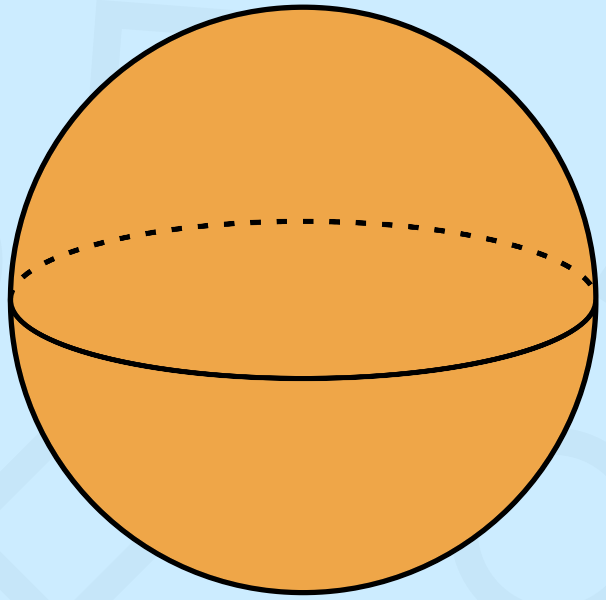
- A cube has 6 faces.
- A cube consists of 6 square faces that are connected together.
- When a cube is unfolded it will give 6 squares.
- All faces are congruent and perpendicular to each other.
- There are 11 nets in a cube.

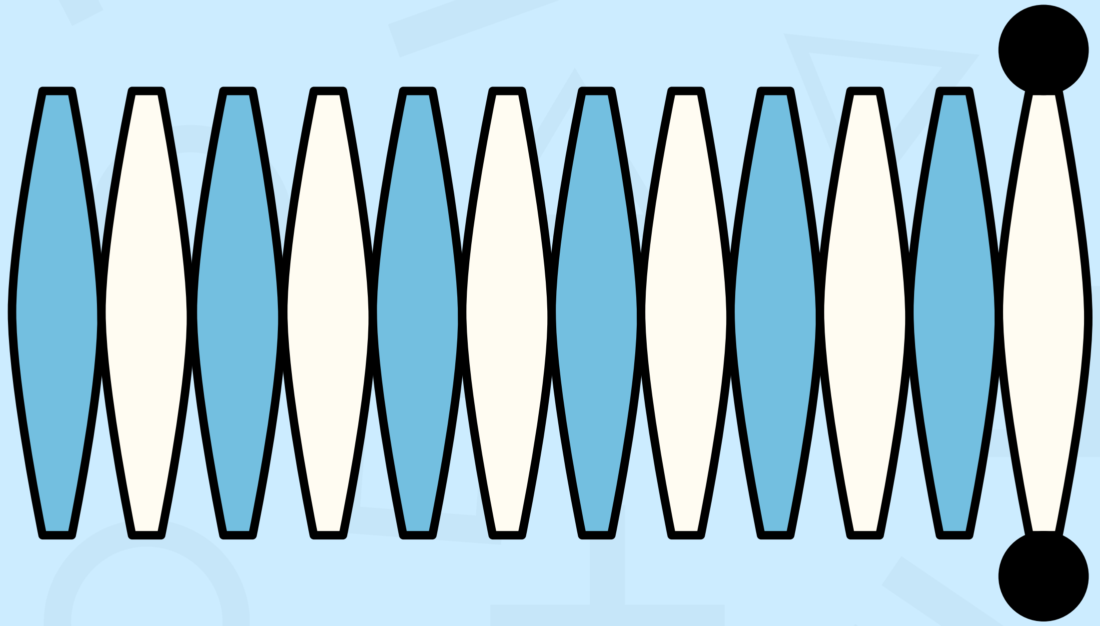




# Net of a Sphere

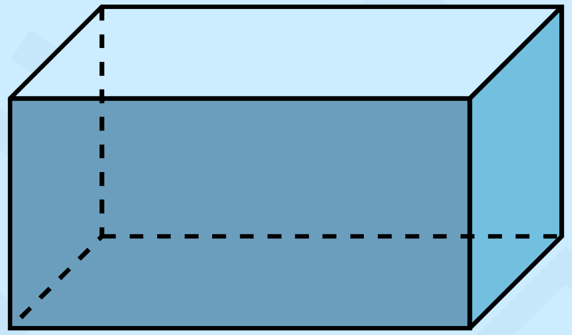
- A sphere is a perfectly symmetrical 3d shape that has no flat surface.
- It has all point equidistant from the center.
- When a sphere is unfolded, it will give a continuous curve.

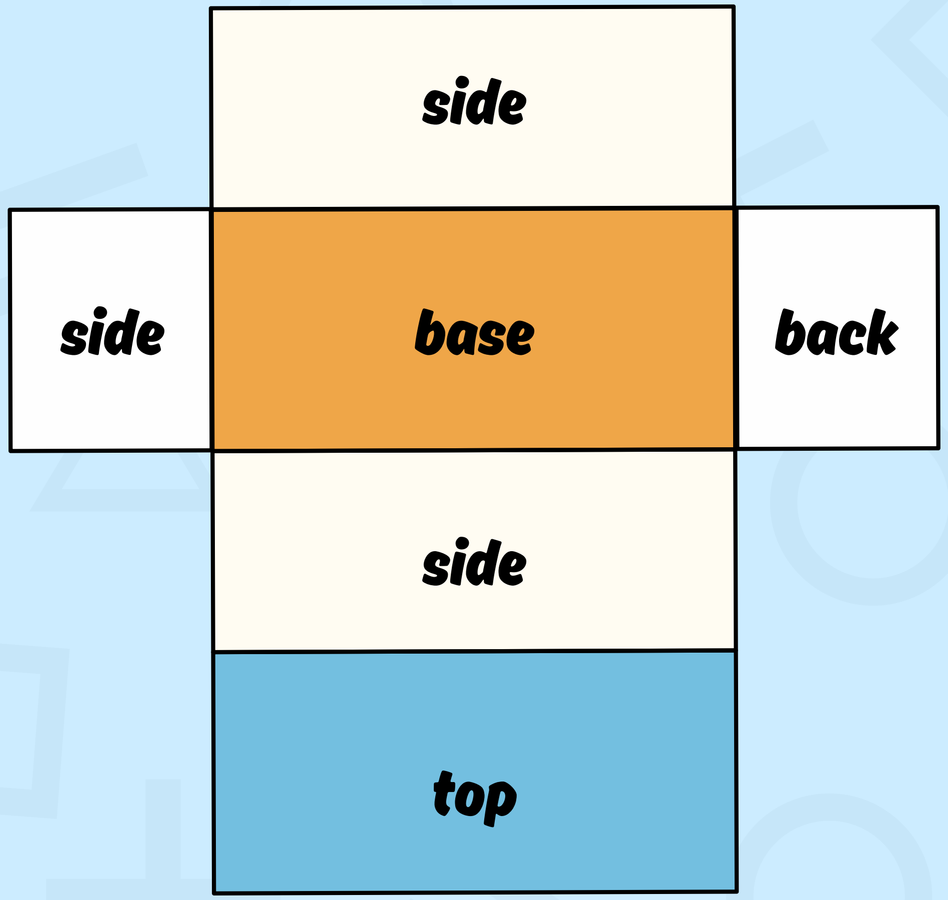




# Net of a Cuboid

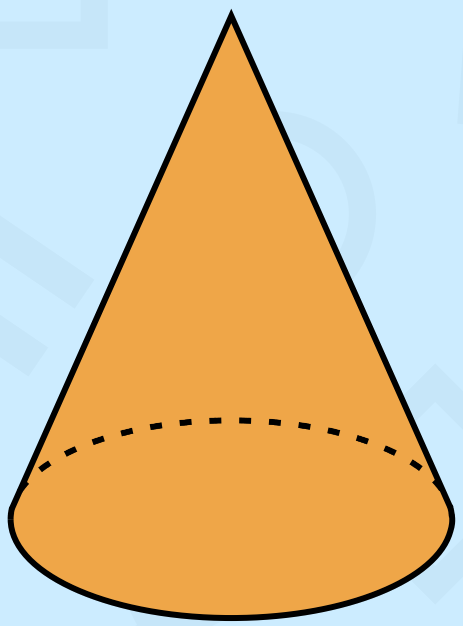
- A cuboid has 6 faces and all of these faces are rectangles
- If a cuboid is unfolded, it will give 6 rectangles.
- Opposite faces are congruent and parallel to each other.
- There are 54 different nets of the cuboid.

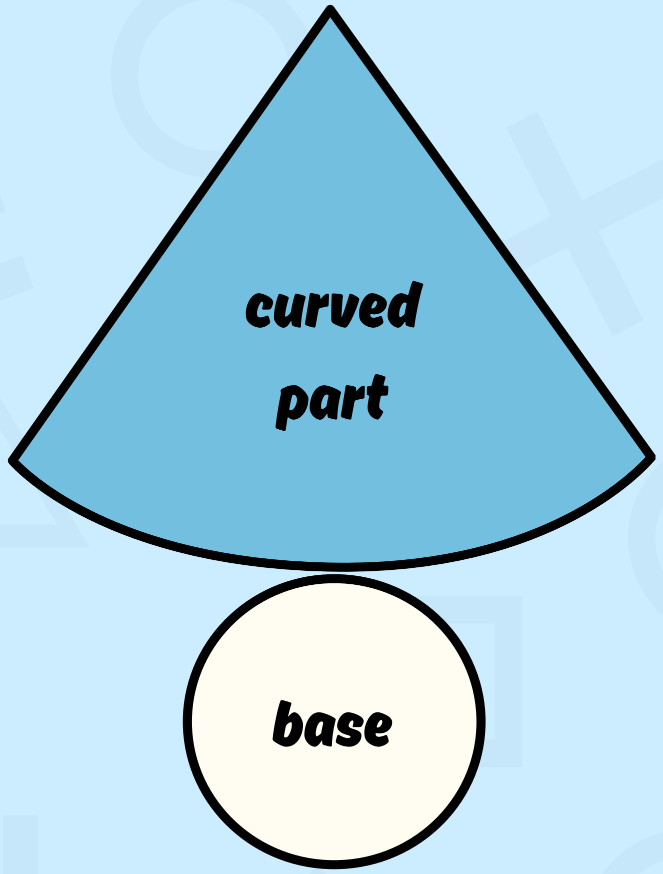




# Net of a Cone

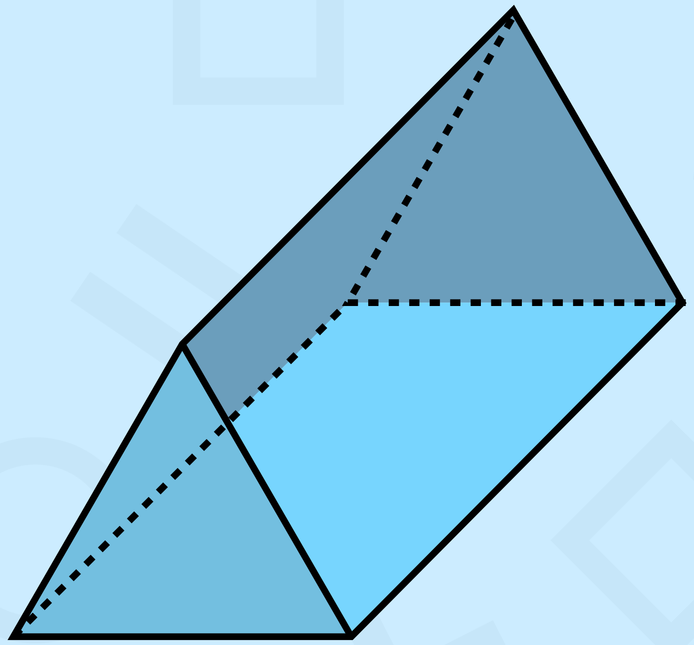
- The cone has only 2 faces.
- Out of the 2 faces, 1 is a curved surface and the other one is a flat face.
- A net of cones has 2 parts: One is the circle that represents the flat part. The second one represents the curved part.
- When it is unfolded, it will give 1 circle and 1 sector that tapers to a single point called apex.

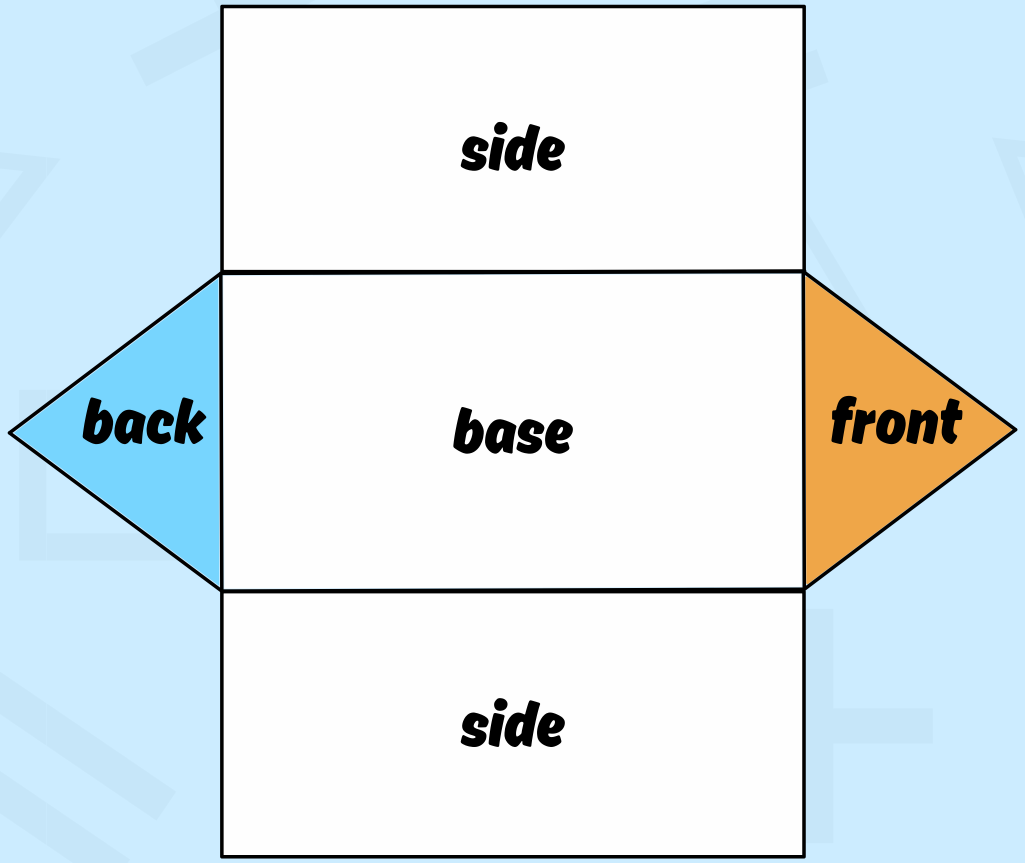




# Net of a Prism

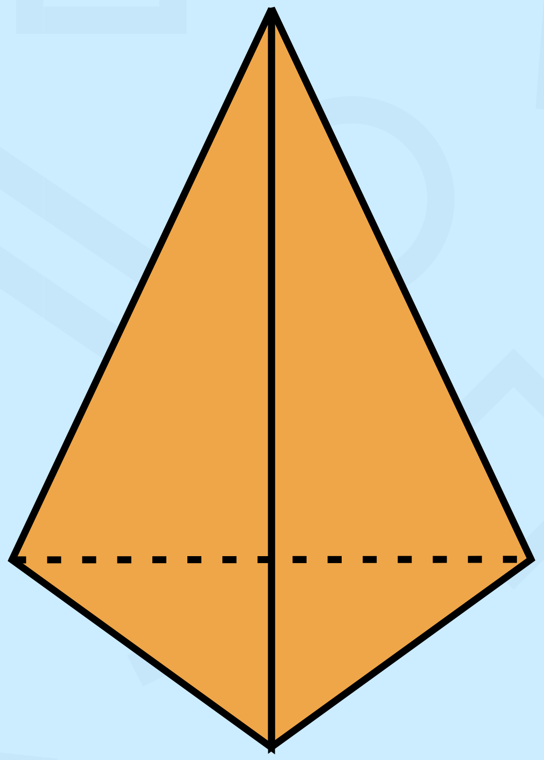
- A prism has 2 identical, congruent polygons that form from the top and bottom of the prism.
- The triangular prism has 2 triangular bases and 3 rectangular lateral faces connecting the corresponding edges of the triangles.

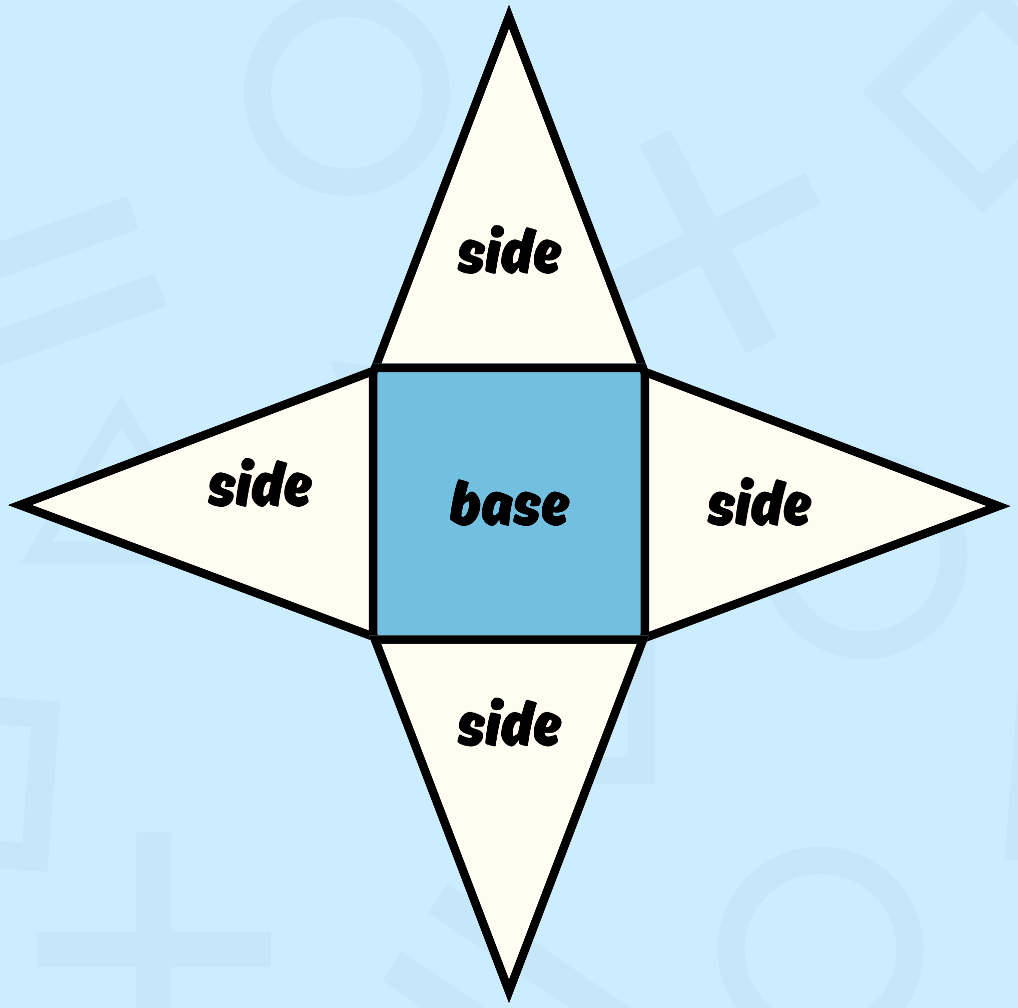




# Net of a Pyramid

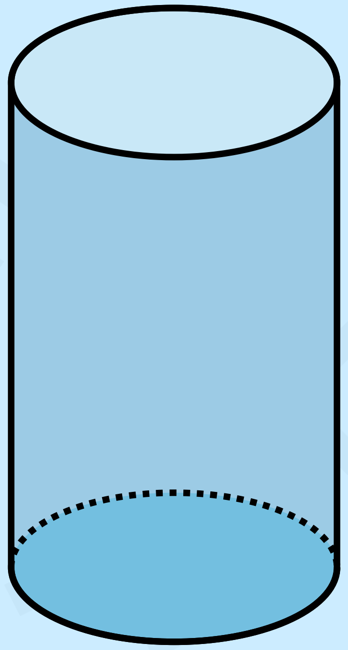
- A pyramid has 5 faces.
- These faces include 1 square and 4 triangular faces.
- When a pyramid is unfolded it will give 1 square and 4 triangles.

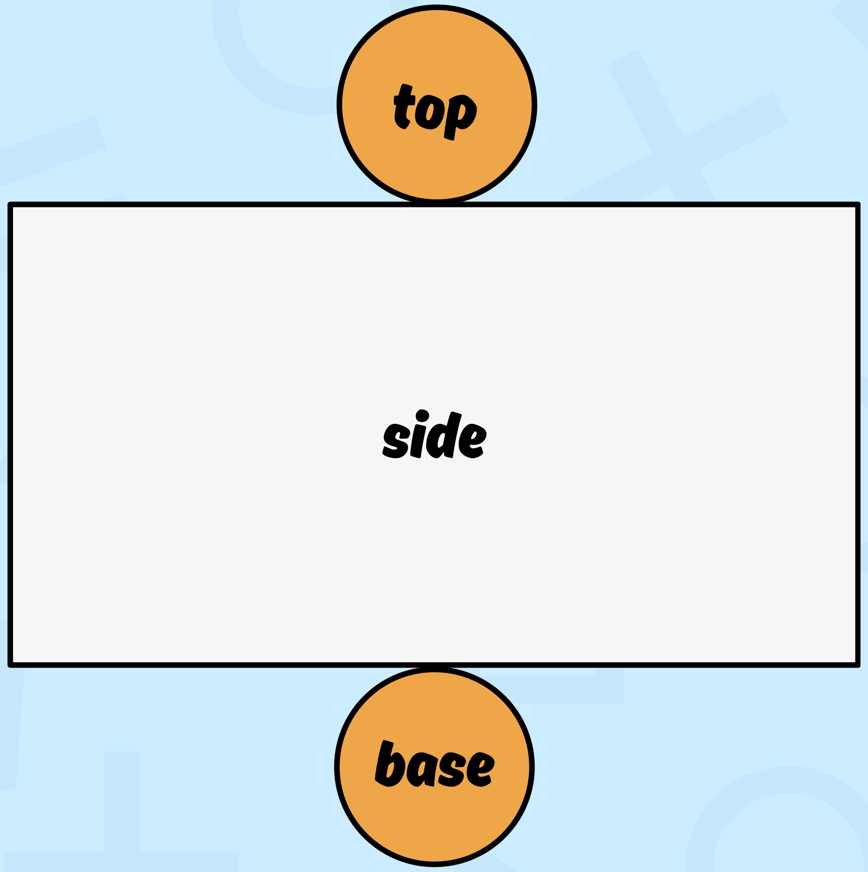




# Net of a Cylinder

- A cylinder is made up of 3 faces.
- Two are circular bases and one is a curved surface.
- Two of the faces are circular and one is rectangular.
- When a cylinder is unfolded, it will give 2 circles and 1 rectangle.





# Thank you

# for listening